C12.5 DVM transactions format

Table C12-2 shows the outline message format and the read address channel address bit encoding for DVM.

Table C12-2 DVM transactions format

ARADDR bits	Name	Function
[(n-1):32]	-	Additional virtual address bits or Reserved, must be zero **Note** n represents the width of the AR address bus
[31:24]	-	*Virtual Machine Identifier* (VMID) or Virtual Index, VA[27:20]
[23:16]	-	*Address Space IDentifier* (ASID) or Virtual Index, VA[19:12]
[15]	Completion	Indicates if a DVM Complete transaction is required: **0** a DVM Complete transaction is not required **1** a DVM Complete transaction is required
[14:12]	Message type	`0b000` TLB Invalidate `0b001` Branch Predictor Invalidate `0b010` Physical Instruction Cache Invalidate `0b011` Virtual Instruction Cache Invalidate `0b100` Synchronization `0b110` Hint
[11:10]	Guest OS or hypervisor	`0b00` Transaction applies to hypervisor and all Guest OS `0b01` Reserved `0b10` Transaction applies to Guest OS `0b11` Transaction applies to hypervisor
[9:8]	Security	`0b00` Transaction applies to Secure and Non-secure `0b01` Reserved `0b10` Transaction applies to Secure only `0b11` Transaction applies to Non-secure only
[7]	-	Reserved, SBZ
[6]	-	Message includes information in **ARADDR[31:24]**
[5]	-	Message includes information in **ARADDR[23:16]**
[4:1]	-	Reserved, SBZ
[0]	-	An additional transaction conveys virtual address information

Table C12-3 shows the format if an additional transaction is used to convey address information.

Table C12-3 DVM additional transactions format

ARADDR bits	Description
[(n-1):4]	Virtual address bits VA[(n-1):4] or Physical address bits PA[(n-1):4]
[3:0]	SBZ

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